

UCCC 2063 Algorithm Analysis

Quiz

Time allowed: 1 hour

Full mark = 75

Q1(a)

- (i) $O(n^7)$
- (ii) $O(n^7)$

Q1(b)

- (i) $O(n^6)$
- (ii) $O(n^2)$
- (iii) $O(n^3)$
- (iv) $O(n^6)$

Q1(c)

- (i) $O(n^6)$
- (ii) $O(\log n)$
- (iii) $O(n \log n)$
- (iv) $O(n^6)$

Q2(a)

$$\exists c = 8, n_0 = 1 \bullet \forall n > n_0 \bullet$$

$$cn^3 > 7n^3$$

$$4n^3 + 3n^3 < cn^3$$

$$4n^3 + 3\sqrt{n} < cn^3$$

Q2(b)

Proof: Need to prove
 $[\exists c, n_0 \bullet \forall n > n_0 \bullet f(n) > cg(n)]$ wrong.

It is equivalent to prove

$$\neg [\exists c, n_0 \bullet \forall n > n_0 \bullet 4n^3 + 3\sqrt{n} > cn^{3.1}]$$

$$\neg [\exists c, n_0 \bullet \forall n > n_0 \bullet 4n^3 + 3\sqrt{n} > cn^{3.1}]$$

$$\equiv \forall c, n_0 \bullet \neg [\forall n > n_0 \bullet 4n^3 + 3\sqrt{n} > cn^{3.1}]$$

$$\equiv \forall c, n_0 \bullet \exists n > n_0 \bullet \neg [4n^3 + 3\sqrt{n} > cn^{3.1}]$$

$$\equiv \forall c, n_0 \bullet \exists n > n_0 \bullet 4n^3 + 3\sqrt{n} \leq cn^{3.1}$$

We now prove

$$\forall c, n_0 \bullet \exists n > n_0 \bullet 4n^3 + 3\sqrt{n} \leq cn^{3.1}.$$

Proof:

$$\forall c, n_0 \bullet \exists n = \max \left(n_0 + 1, \left\lceil \left(\frac{7}{c} \right)^{10} \right\rceil \right) > n_0 \bullet$$

$$n \geq \left(\frac{7}{c} \right)^{10}$$

$$cn^{0.1} \geq 7$$

$$7n^3 \leq cn^{3.1}$$

$$4n^3 + 3n^3 \leq cn^{3.1}$$

$$4n^3 + 3\sqrt{n} \leq cn^{3.1}.$$

Q2(c)

Assume $O(n^2) \cap \Omega(n^5) \neq \emptyset$, for contradiction.

Let $f(n) \in O(n^2) \cap \Omega(n^5)$.

Then, $f(n) = O(n^2)$ and $f(n) = \Omega(n^5)$.

So, $[\exists c_0, n_0 \bullet \forall n > n_0 \bullet f(n) < c_0 n^2]$ and
 $[\exists c_1, n_1 \bullet \forall n > n_1 \bullet f(n) > c_1 n^5]$

Combining the inequalities into one,

$$\exists c_0, n_0, c_1, n_1 \bullet \forall n > \max(n_0, n_1) \bullet$$

$$c_1 n^5 < f(n) < c_0 n^2$$

$$c_1 n^5 < c_0 n^2$$

$$n < \left(\frac{c_0}{c_1} \right)^{1/3}$$

Contradiction is reached $\forall n > \max \left(n_0, n_1, \left(\frac{c_0}{c_1} \right)^{1/3} \right)$.

Q2(d)

$\log n!$

$$= \log 1 + \log 2 + \dots + \log n$$

$$> \log(\lfloor n/2 \rfloor + 1) + \log(\lfloor n/2 \rfloor + 2) + \dots + \log n$$

$$> \log(\lfloor n/2 \rfloor + 1) + \log(\lfloor n/2 \rfloor + 1) + \dots + \log(\lfloor n/2 \rfloor + 1)$$

$$< \text{-----} (n - \lfloor n/2 \rfloor) \text{ terms} \text{-----}>$$

$$= (n - \lfloor n/2 \rfloor) \log(\lfloor n/2 \rfloor + 1)$$

$$= (n/2) \log(n/2)$$

$$= (1/2) n \log n - (1/2)n$$

$$= \Omega(n \log n)$$